Characteristic Classes

Book Notes of *Characteristic Classes*

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1 Obstruction Theory & Characteristic Classes

1.1 A Brief Introduction to Obstruction Theory

In homotopy theory, one often encounters the following problem:

> Given a pair of spaces (X, A) and a map $f : A \to Y$. Can f be extended to a map $F : X \to Y$?

Obstruction theory provides a systematic way to study this problem.

Toy Model : A map $f : S^n \to Y$ can be extended to a map $F : D^{n+1} \to Y$ if and only if f is null-homotopic.

Now we consider the general case. Let K be a CW complex, Y be a simple space, i.e., $\pi_1(Y)$ acts trivially on $\pi_n(Y)$ for all n. Under this condition, we can unambiguously use the notation $\pi_n(Y) = [S^n, Y]$. Suppose we have a map $f : K^{(n)} \to Y$, where $K^{(n)}$ is the n-skeleton of K. We want to extend f to a map $F : K^{(n+1)} \to Y$. To do this, we only need to extend f to each $(n+1)$-cell e_α^{n+1}.

Suppose the attaching map of e_α^n is $\varphi_\alpha : S^n \to K^{(n)}$. Then $f \circ \varphi_\alpha$ is a map from S^n to Y. By the toy model, $f \circ \varphi_\alpha$ can be extended to D^{n+1} if and only if $[f \circ \varphi_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^{n+1}(f) \in C_{cell}^{n+1}(K; \pi_n(Y))$:

$$\begin{aligned} c^{n+1}(f) : C_{n+1}^{cell}(K) &\to \pi_n(Y) \\ e_\alpha^{n+1} &\mapsto [f \circ \varphi_\alpha]. \end{aligned}$$

We call $c^{n+1}(f)$ the $(n+1)$**-th obstruction cochain** of f. There are some basic facts about $c^{n+1}(f)$:

- f can be extended to $K^{(n+1)}$ if and only if $c^{n+1}(f) = 0$;
- If $f, f' : K^{(n)} \to Y$ are homotopic, then $c^{n+1}(f) = c^{n+1}(f')$;
- (Natruality) Suppose L is another CW complex and $\phi : L \to K$ is a cellular map, then $c^{n+1}(f \circ \phi) = \phi^\sharp c^{n+1}(f)$.

At the following, we give some important properties of $c^{n+1}(f)$ without proof.

Theorem 1.1. *$c^{n+1}(f)$ is a cocycle, i.e., $\delta c^{n+1}(f) = 0$. Thus it defines a cohomology class $[c^{n+1}(f)] \in H^{n+1}(K; \pi_n(Y))$. We call $[c^{n+1}(f)]$ the* $(n+1)$***-th obstruction class*** *of f.*

A similar concept is defined for homotopies. Suppose we have two maps $f_0, f_1 : K^{(n)} \to Y$ and a homotopy $h : K^{(n-1)} \times I \to Y$ between $f_0|_{K^{(n-1)}}$ and $f_1|_{K^{(n-1)}}$. We want to extend h to a homotopy $H : K^{(n)} \times I \to Y$ between f_0 and f_1. To do this, we only need to extend h to each n-cell e^n_α. Let $\varphi_\alpha : S^{n-1} \to K^{(n-1)}$ be the attaching map of e^n_α, $\psi_\alpha : D^n \to K^{(n)}$ be the characteristic map of e^n_α. Then we can form a map

$$\begin{aligned} h_\alpha : (D^n \times \{0{,}1\}) \cup (\partial D^n \times I) &\to Y \\ (x, 0) &\mapsto f_0 \circ \psi_\alpha(x) \\ (x, 1) &\mapsto f_1 \circ \psi_\alpha(x) \\ (x, t) &\mapsto h \circ (\varphi_\alpha(x), t). \end{aligned}$$

Since $(D^n \times \{0, 1\}) \cup (\partial D^n \times I) \cong S^n$, h_α defines a map from S^n to Y. It can extend to $D^n \times I \cong D^{n+1}$ if and only if $[h_\alpha] = 0$ in $\pi_n(Y)$. We in fact get a cochain $c^n(f_0, f_1, h) \in C^n_{cell}(K; \pi_n(Y))$:

$$\begin{aligned} c^n(f_0, f_1, h) : C_n^{cell}(K) &\to \pi_n(Y) \\ e^n_\alpha &\mapsto [h_\alpha]. \end{aligned}$$

We call $c^n(f_0, f_1, h)$ the **n-th deformation cochain** of (f_0, f_1, h). If $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}}$ and h is the trivial homotopy, we simply write $c^n(f_0, f_1)$ for $c^n(f_0, f_1, h)$ and call it the **n-th difference cochain** of (f_0, f_1). There are some basic facts about $c^n(f_0, f_1, h)$:

- h can be extended to a homotopy $H : K^{(n)} \times I \to Y$ between f_0 and f_1 if and only if $c^n(f_0, f_1, h) = 0$;
- (Natruality) Suppose L is another CW complex and $\phi : L \to K$ is a cellular map, then $c^n(f_0 \circ \phi, f_1 \circ \phi, h \circ (\phi \times \mathrm{id}_I)) = \phi^\sharp c^n(f_0, f_1, h)$.
- $c^n(f_0, f_2, h * h') = c^n(f_0, f_1, h) + c^n(f_1, f_2, h')$, where $h * h'$ is the concatenation of h and h'. Especially, if $f_0|_{K^{(n-1)}} = f_1|_{K^{(n-1)}} = f_2|_{K^{(n-1)}}$ and h, h' are trivial homotopies, then $c^n(f_0, f_2) = c^n(f_0, f_1) + c^n(f_1, f_2)$.

In general, $c^n(f_0, f_1, h)$ is not a cocycle. But we have the following formula.

Theorem 1.2. $\delta c^n(f_0, f_1, h) = c^{n+1}(f_1) - c^{n+1}(f_0)$.

There is a useful lemma about difference cochains.

Lemma 1.3. *For any $f : K^{(n)} \to Y$ and any cochain $d \in C^n_{cell}(K; \pi_n(Y))$, there exists a map $f' : K^{(n)} \to Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$ and $d^n(f, f') = d$.*

With the above preparations, we have the following theorem:

Theorem 1.4 (Eilenberg)**.** *Given a map $f : K^{(n)} \to Y$, $[c^n(f)] = 0$ in $H^{n+1}(K; \pi_n(Y))$ if and only if f can extend to $K^{(n+1)}$ after having been changed on $K^{(n)} \backslash K^{(n-1)}$.*

Proof. $(\Leftarrow)$: If $\exists\, f' : K^{(n+1)} \to Y$ such that $f'|_{K^{(n-1)}} = f|_{K^{(n-1)}}$, then

$$c^{n+1}(f) = c^{n+1}(f) - c^{n+1}(f') = \delta d^n(f', f)$$

Thus $c^{n+1}(f)$ is a coboundary.

$(\Rightarrow)$: If $[c^{n+1}(f)] = 0$, then $\exists\, d \in C^n(K; \pi_n(Y))$ such that $c^{n+1}(f) = \delta d$. By lemma, there exists $f' : K^{(n)} \to Y$ such that $f|_{K^{(n-1)}} = f'|_{K^{(n-1)}}$ and $d^n(f, f') = -d$. So

$$c^{n+1}(f') = c^{n+1}(f) + \delta d^n(f, f') = \delta d - \delta d = 0.$$

Thus f' can be extended to $K^{(n+1)}$. $\square$

Example 1.5. If the target space Y is n-connected, i.e., $\pi_k(Y) = 0,\ k = 0, 1, \ldots, n$, then every map $f : K^{(0)} \to Y$ can be extended to $K^{(n+1)}$. Besides, any two maps $f, g : K^{(n)} \to Y$ are homotopic. Indeed, $f|_{K^{(0)}} \simeq g|_{K^{(0)}}$ for the reason that K is path-connected. We denote the homotopy by h_0. Then h_0 can be extened to $h_n : K^{(n)} \times I \to Y$ since all the deformation cochain vanish.

Corollary 1.6. *If Y is n-connected and $\dim K \leq n$, then all the maps from K to Y are null-homotopic.*

Theorem 1.7 (Hopf-Whitehead)**.** *For any CW complex K with $\dim K \leq n$ and $(n-1)$-connected space Y, there is a one-to-one correspondence:*

$$[K, Y] \longleftrightarrow H^n(K; \pi_n(Y)).$$

If we weaken the dimensional condition and instead strengthen the target space to be an Eilenberg-Maclane space $K(\pi, n)$. Then we will get another theorem:

Theorem 1.8. *For any CW complex X, abelian group π and integer n, there is a one-to-one correspondence:*

$$[X, K(\pi, n)] \longleftrightarrow H^n(X; \pi).$$

Remark. In fact, there is a special class α in

$$H^n(K(\pi,n);\pi) \cong \mathrm{Hom}(H_n(K(\pi,n)),\pi) \cong \mathrm{Hom}(\pi,\pi)$$

which corresponds to the identity map in $\mathrm{Hom}(\pi,\pi)$. The correspondence can be explicitly written by

$$\begin{aligned}[X, K(\pi,n)] &\to H^n(X;\pi)\\ f &\mapsto f^*\alpha.\end{aligned}$$

1.2 Characteristic Classes via Obsruction Theory

1.2.1 Cohomology with Local Coefficients

To discrib characteristic classes as obstruction classes, we need cohomology with local coefficients.

Local Coefficient System:

Suppose X is a path-connected space. For any $x \in X$, let G_x be an abelian group attaching to x. For any path γ_{xy} from x to y, we have an isomorphism $(\gamma_{xy})_* : G_x \to G_y$. If $(\gamma_{xy})_*$ only depends on the homotopy class of γ rel. endpoints and satisfies the following conditions:

$$(\text{constant path})_* = \mathrm{Id}, \quad (\gamma_{xy} * \gamma_{yz})_* = (\gamma_{yz})_* \circ (\gamma_{xy})_*.$$

Then we call $\mathcal{G} = \{G_x\}_{x\in X}$ a **local coefficient system** on X. If we fix a base point $x_0 \in X$, then $\mathcal{G}$ is determined by the group $G = G_{x_0}$ and an action $\rho : \pi_1(X, x_0) \to \mathrm{Aut}(G)$. We call G the **typical group** of $\mathcal{G}$.

Example 1.9. If $\mathcal{G}$ is a local coefficient system on X with typical group G and trivial action. Then it is the ordinary constant coefficient we are familiar with.

Example 1.10. If $G_x = \mathbb{Z}_2$ for all $x \in X$, then since $\mathrm{Aut}(\mathbb{Z}_2) = \{\mathrm{Id}\}$, the local coefficient system with typical group $\mathbb{Z}_2$ must be trivial.

Example 1.11. If $G_x = \mathbb{Z}$ for all $x \in X$ and $\pi_1(X, x_0)$ acts on $\mathbb{Z}$ by multiplication of ± 1 according to whether the orientation is preserved or reversed along the loop. Then $\mathcal{G}$ is called the **orientation system** on X and denoted by Or.

Example 1.12. Let $p : \tilde{X} \to X$ be the universal covering of X. Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x\in X}$ is a local coefficient system on X with typical group $H_n(\tilde{X})$ and action induced by the deck transformation group $\pi_1(X, x_0)$.

Example 1.13. Let $E \to X$ be a vector bundle with fiber F. Then $\mathcal{G} = \{H_n(p^{-1}(x))\}_{x \in X}$ is a local coefficient system on X with typical group $H_n(F)$ and action induced by the structure group of E.

Example 1.14. $\mathcal{G} = \{\pi_n(X, x)\}_{x \in X}$ is a local coefficient system on X with typical group $\pi_n(X, x_0)$ and action induced by the fundamental group $\pi_1(X, x_0)$.

Cohomology with Local Coefficients:

Let X be a CW complex and $\mathcal{G} = \{G_x\}_{x \in X}$ be a local coefficient system on X. We define the cellular cochain groups with local coefficients by

$$C^n_{cell}(X; \mathcal{G}) = \left\{ f : \{n\text{-cells of } X\} \to \coprod_{x \in X} G_x \,\middle|\, f(e^n_\alpha) \in G_{x_\alpha} \right\}$$

where x_α is a chosen point in the n-cell e^n_α. The coboundary operator $\delta : C^n_{cell}(X; \mathcal{G}) \to C^{n+1}_{cell}(X; \mathcal{G})$ is defined by

$$(\delta f)(e^{n+1}_\beta) = \sum_\alpha d_{\beta\alpha} (\gamma_{\alpha\beta})_* f(e^n_\alpha),$$

where $d_{\beta\alpha}$ is the degree of the map $\partial e^{n+1}_\beta \to X^{(n)}/(X^{(n)} - e^n_\alpha) \cong S^n$ and $\gamma_{\alpha\beta}$ is a path from x_α to x_β. It can be verified that $\delta^2 = 0$. Thus we can define the cohomology groups with local coefficients by

$$H^n(X; \mathcal{G}) = \frac{\ker(\delta : C^n_{cell}(X; \mathcal{G}) \to C^{n+1}_{cell}(X; \mathcal{G}))}{\operatorname{im}(\delta : C^{n-1}_{cell}(X; \mathcal{G}) \to C^n_{cell}(X; \mathcal{G}))}.$$

The Homotopy Groups of $V_k(\mathbb{R}^n)$

Fact: The Stiefel manifold $V_k(\mathbb{R}^n)$ is $(n-k-1)$-connected and the first nontrivial homotopy group is

$$\pi_{n-k}(V_k(\mathbb{R}^n)) = \begin{cases} \mathbb{Z}, & n-k \text{ is even or } k = 1; \\ \mathbb{Z}_2, & n-k \text{ is odd and } k > 1. \end{cases}$$

Remark. It is evident that $V_k(\mathbb{R}^n)$ is a simple space.

Characteristic Classes

Now let's back to vector bundles. Let ξ be an $\mathbb{R}^n$-bundle over a CW complex B. We want to investigate how nontrivial ξ is. To do this, we try to find k-linearly independent sections of ξ.

Let $V_k(\xi)$ be the Stiefel manifold bundle associated to ξ. It's a fiber bundle over B with fiber $F = V_k(\mathbb{R}^n)$. A section s of $V_k(\xi)$ is a k-frame of ξ. Locally on a trivializing neighborhood U_α, s can be identified as a map from U_α to $V_k(\mathbb{R}^n)$. Thus we can use obsruction theory to study the existence of sections of $V_k(\xi)$.

The obstructions $c^{j+1}(s)$ lie in $H^{j+1}(B; \mathcal{G})$, where $\mathcal{G}$ is a local coefficient system on B with typical group $\pi_j(V_k(\mathbb{R}^n))$. This is because although the ristriction of $V_k(\xi)$ on every j-cell e^j_α is trivial, the transition function may not be identity, from which induces a nontrivial isomorphism of homotopy groups. Moreover, any path γ_{xy} from x to y can lift to a homotopy from F_x to F_y, thus induces an isomorphism

$$(\gamma_{xy})_* : \pi_j\big(F_x, x\big) \overset{\cong}{\to} \pi_j\big(F_y, y\big).$$

The theorems about obstruction classes can all generalize to cohomology with local coefficients without difficulty.

Now since $V_k(\mathbb{R}^n)$ is $(n-k-1)$-connected, all the section s can be extended to $B^{(n-k)}$, all the sections on $B^{(n-k)}$ are homotopic with each other. The first obstruction comes when extend s to $(n-k+1)$-skeleton, which we denote by

$$\mathfrak{o}_{n-k+1} := c^{n-k+1}(s) \in H^{n-k+1}\Big(B; \big\{\pi_{n-k}\big(V_k(\mathbb{R}^n)\big)\big\}\Big)$$

Since all sections on $B^{(n-k)}$ are homotopic with each other, this obstruction class does not depend on the choice of s, i.e, $\mathfrak{o}_{n-k+1}$ it is well-defined.

Now we construct a series of characteristic classes from obsruction classes.

- If $n-k = 2j-1$ is odd and $1 < k < n$, then $\pi_{2j-1}(V_{n-2j+1}(\mathbb{R}^n)) = \mathbb{Z}_2$. In this case,

$$\begin{aligned} H^{2j}\Big(B; \big\{\pi_{2j-1}\big(V_{n-2j+1}(\mathbb{R}^n)\big)\big\}\Big) &\cong H^{2j}(B; \mathbb{Z}_2) \\ \mathfrak{o}_{2j} &\mapsto w_{2j}(\xi). \end{aligned}$$

- If $n-k = 2j$ is even and $1 < k < n$, then $\pi_{2j}(V_{n-2j}(\mathbb{R}^n)) = \mathbb{Z}$. After applying the mod 2 reduction, we have

$$\begin{aligned} H^{2j+1}\Big(B; \big\{\pi_{2j}\big(V_{n-2j}(\mathbb{R}^n)\big)\big\}\Big) &\xrightarrow{\text{mod } 2} H^{2j+1}(B; \mathbb{Z}_2) \\ \mathfrak{o}_{2j+1} &\mapsto w_{2j+1}(\xi). \end{aligned}$$

- If $k = n$, then $V_n(\mathbb{R}^n) = O(n)$, but $\pi_0(O(n)) = \mathbb{Z}_2$ is not a group. We turn to use $\tilde{H}^0(O(n); \mathbb{Z}) = \mathbb{Z}$ instead. Then $\tilde{H}^0(O(n); \mathbb{Z})$ is a local coefficient system on B with typical group $\mathbb{Z}$ and action induced by the determinant representation $\det : \pi_1(B) \to O(n) \to \{\pm 1\}$. After applying the mod 2 reduction, we have

$$H^1\Big(B; \{\tilde{H}^0(O(n))\}\Big) \xrightarrow{\text{mod } 2} H^1(B; \mathbb{Z}_2)$$
$$\mathfrak{o}_1 \mapsto w_1(\xi).$$

- If $k = 1$, then $\pi_{n-1}(V_1(\mathbb{R}^n)) = \pi_{n-1}(S^{n-1}) = \mathbb{Z}$. After applying the mod 2 reduction, we have

$$H^n\Big(B; \{\pi_{n-1}(V_1(\mathbb{R}^n))\}\Big) \xrightarrow{\text{mod } 2} H^n(B; \mathbb{Z}_2)$$
$$\mathfrak{o}_n \mapsto w_n(\xi).$$

Cases	Obstruction classes	Characteristic classes
$n-k = 2j-1,\ 1 < k < n$	$\mathfrak{o}_{2j} = w_{2j}$	
$n-k = 2j,\ 1 < k < n$	$\mathfrak{o}_{2j+1} \xrightarrow{\text{mod } 2} w_{2j+1}$	
$k = n$	$\mathfrak{o}_1 \xrightarrow{\text{mod } 2} w_1$	
$k = 1$	$\mathfrak{o}_n \xrightarrow{\text{mod } 2} w_n$	

Properties of our construction

Stiefel-Whitney Classes

Theorem 1.15. *$w_i(\xi) \in H^i(B; \mathbb{Z}_2)$ are the Stiefel-Whitney classes of ξ.*

Proof. omitted. $\square$

$\mathfrak{o}_n$ and the Euler Class If ξ is orientable, then the structure group of ξ can be reduced to $SO(n)$. Thus the action of $\pi_1(B)$ on $\pi_{n-k}(V_k(\mathbb{R}^n))$ is trivial for $1 \leq k < n$. So the local coefficient system is trivial. In this case,

$$H^n\Big(B; \{\pi_{n-1}(V_1(\mathbb{R}^n))\}\Big) \cong H^n(B; \mathbb{Z}).$$

Theorem 1.16. *If ξ is orientable, then $\mathfrak{o}_n \in H^n(B; \mathbb{Z})$ is the Euler class $e(\xi)$.*

Proof. omitted. $\square$

w_1 **and Orientability** The structure group of ξ can be reduced to $SO(n)$ if and only if the transition functions all have determinant 1, i.e., the action of $\pi_1(B)$ on $\tilde{H}^0(O(n);\mathbb{Z})$ is trivial. This is equivalent to the triviality of $\mathfrak{o}_1$. Thus

Proposition 1.17. $\mathfrak{o}_1 = 0$ *if and only if* ξ *is orientable.*

Theorem 1.18. $\mathfrak{o}_1$ *has order 2, i.e.,* $2\mathfrak{o}_1 = 0$*. Thus*

$$w_1(\xi) = 0 \text{ if and only if } \mathfrak{o}_1 = 0.$$

Proof. The connected conponents of the space $V_n(\mathbb{R}^n)$ are identified with orientation of the fiber, and hence the elements of $\tilde{H}^0(V_n(\mathbb{R}^n);\mathbb{Z}) \cong \mathbb{Z}$ has the form $a\mathrm{Or}_1 - a\mathrm{Or}_2$; to each orientation we assign its Coefficient.

... □

Is w_i the true obsruction? The fibration

$$S^{2j-1} \to V_{n-2j+1}(\mathbb{R}^n) \to V_{n-2j}(\mathbb{R}^n).$$

induces an exact sequence of homotopy groups

$$\pi_{2j}(V_{n-2j}(\mathbb{R}^n)) \xrightarrow{\partial} \pi_{2j-1}(S^{2j-1}) \to \pi_{2j-1}(V_{n-2j+1}(\mathbb{R}^n)) \to 0.$$

It is actually

$$\mathbb{Z} \xrightarrow{\partial} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}_2 \to 0.$$

Therefore $\mathrm{Im}\,\partial = \mathrm{Ker} i_* = 2\mathbb{Z}$ and $\mathrm{Ker}\partial = 0$. So ∂ is multiplication by 2. So the exact sequence is a short exact sequence

$$0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \xrightarrow{i_*} \mathbb{Z}_2 \to 0.$$

This short exact sequence induces a long exact sequence of cohomology groups

$$\cdots \to H^{2j}(B;\mathbb{Z}) \xrightarrow{i_*} H^{2j}(B;\mathbb{Z}_2) \xrightarrow{\beta} H^{2j+1}(B;\mathbb{Z}) \xrightarrow{\times 2} H^{2j+1}(B;\mathbb{Z}) \to \cdots$$

where β is the Bockstein homomorphism.

Theorem 1.19. $\mathfrak{o}_{2j+1} = \beta(\mathfrak{o}_{2j})$.

Proof. omitted. □

Corollary 1.20. $\mathfrak{o}_{2j+1}$ *has order 2, i.e.,* $2\mathfrak{o}_{2j+1} = 0$*. Thus*

$$w_{2j+1}(\xi) = 0 \text{ if and only if } \mathfrak{o}_{2j+1} = 0.$$

In summary, almost all the Stiefel-Whitney classes w_i are the true obstructions, except when $i = n$ is even. If in addition ξ is oriented, then $\mathfrak{o}_n$ is Euler class $e(\xi)$.

2 Chern Numbers & Pontrjagin Numbers

2.1 Chern Numbers

Definition 2.1. Let K^n be a compact complex manifold, $\dim_{\mathbb{C}} K = n$. For each partition $I = i_1, \dots, i_r$ of n, we define the I-th Chern number to be the integer

$$c_I[K^n] = c_{i_1} \cdots c_{i_r}[K^n] := \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{K^n} \rangle,$$

where μ_{K^n} is the fundamental homology class w.r.t. the orientation determined by the complex structure.

If I is a partition of other number, we define $c_I[K^n]$ to be 0.

Example 2.2. For $K^n = \mathbb{C}P^n$, we have

$$\begin{aligned} c_I[\mathbb{C}P^n] &= \langle c_{i_1}(\tau) \cdots c_{i_r}(\tau), \mu_{\mathbb{C}P^n} \rangle \\ &= \left\langle \binom{n+1}{i_1} \cdots \binom{n+1}{i_r} a^n, \mu_{\mathbb{C}P^n} \right\rangle \\ &= \binom{n+1}{i_1} \cdots \binom{n+1}{i_r}. \end{aligned}$$

2.2 Pontrjagin Numbers

Definition 2.3. Let M^{4n} be a compact oriented smooth manifold of dimension $4n$. For each partition $I = i_1, \dots, i_r$ of n, we define the I-th Pontrjagin number to be the integer

$$p_I[M^{4n}] = p_{i_1} \cdots p_{i_r}[M^{4n}] := \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{M^{4n}} \rangle,$$

where $\mu_{M^{4n}}$ is the fundamental homology class w.r.t. the given orientation.

If I is a partition of other number, we define $p_I[M^{4n}]$ to be 0.

Example 2.4. For $M^{4n} = \mathbb{C}P^{2n}$, we have

$$\begin{aligned} p_I[\mathbb{C}P^{2n}] &= \langle p_{i_1}(\tau) \cdots p_{i_r}(\tau), \mu_{\mathbb{C}P^{2n}} \rangle \\ &= \left\langle \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r} a^{2n}, \mu_{\mathbb{C}P^{2n}} \right\rangle \\ &= \binom{2n+1}{i_1} \cdots \binom{2n+1}{i_r}. \end{aligned}$$

Oriented reversing map. If we reverse the orientation of M^{4n}, since the definition of Pontrjagin classes do not depend on the orientation, we have the following

$$p_k(M) \mapsto p_k(M),$$

$$\mu_M \mapsto -\mu_M,$$
$$p_I[M] \mapsto -p_I[M].$$

Thus the sign of Pontrjagin numbers change when reversing the orientation of M. If M has a nonzero Pontrjagin number, M cannot possess any orientation reversing diffeomorphism.

Manifold as a boundary. If M^{4n} is a boundary of some smooth, compact, oriented $(4n+1)$-dimensional manifold-with-boundary. Then all Pontrjagin numbers of M must be zero. Thus if some $p_I[M] \neq 0$, then M cannot be the boundary of such manifold.

2.3 Symmetric Functions

We know that all symmetric polynomials in $\mathbb{Z}[t_1, \ldots, t_n]$ is itself a polynomial ring:

$$\mathcal{S} = \mathbb{Z}[\sigma_1, \ldots, \sigma_n] \subset \mathbb{Z}[t_1, \ldots, t_n],$$

where σ_i is the i-th elementary symmetric polynomial.

$$1 + \sigma_1 + \cdots + \sigma_n = (1 + t_1) \cdots (1 + t_n).$$

If we focus on the symmetric polynomials of degree k, denoted by $\mathcal{S}^k$, clearly it is a free abelian group with basis

$$\sigma_{i_1} \cdots \sigma_{i_r}, \quad I = i_1, \ldots, i_r \text{ is a partition of } k.$$

For some reason, it is more convenient to use another basis of $\mathcal{S}^k$, which is described as follows.

Define two monomials in $\mathbb{Z}[t_1, \ldots, t_n]$ to be equivalent if they differ by a permutation of the variables. We define

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

to be the summation of all distinct monomials equivalent to $t_1^{a_1} \cdots t_n^{a_n}$. Clearly, it is a symmetric polynomial.

Lemma 2.5.

$$\sum t_1^{a_1} \cdots t_r^{a_r}$$

where $a_1 \cdots a_r$ ranges over all partitions of k with length $r \leq n$, form another basis of $\mathcal{S}^k$.

Now for each partition $I = i_1, \ldots, i_r$ of k, we define a polynomial s_I in k variables as follows: choose $n \geq k$, define

$$s_I(\sigma_1, \ldots, \sigma_k) := \sum t_1^{i_1} \cdots t_r^{i_r},$$

both sides are symmetric polynomials in n variables.

Notes that s_I does not depend on the choice of n, for if we choose another $n' > n$, then we can set $t_{n+1} = \cdots = t_{n'} = 0$ to get back to the previous case.

Clearly, all s_I form a basis of $\mathcal{S}^k$ as I ranges over all partitions of k.

Now let's look at the cohomology ring $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$. the splitting map

$$\phi : \mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty \to \mathrm{Gr}_n(\mathbb{C}^\infty)$$

induces an injective homomorphism of cohomology rings

$$\phi^* : H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) \to H^*(\mathbb{C}P^\infty \times \cdots \times \mathbb{C}P^\infty; \mathbb{Z}) \cong \mathbb{Z}[t_1, \ldots, t_n],$$

where t_i is the generator of the i-th factor. ϕ^* maps $H^*(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z}) = \mathbb{Z}[c_1, \ldots, c_n]$ isomorphically onto the symmetric polynomial subring $\mathcal{S} \subset \mathbb{Z}[t_1, \ldots, t_n]$, with c_i mapped to σ_i.

Since $\phi^* s_I(c_1, \ldots, c_k) = s_I(\sigma_1, \ldots, \sigma_k)$ forms a basis of $\mathcal{S}^k$ as I ranges over all partitions of k, by the injectivity of ϕ^*, $s_I(c_1, \ldots, c_k)$ also forms a basis of $H^{2k}(\mathrm{Gr}_n(\mathbb{C}^\infty); \mathbb{Z})$.

2.4 A Product Formula

Let ω be an n-dimensional complex vector bundle over a compact complex manifold B. For each $k \geq 0$ and partition $I = i_1, \ldots, i_r$ of k, we define

$$s_I(c(\omega)) := s_I(c_1(\omega), \ldots, c_k(\omega)) \in H^{2k}(B; \mathbb{Z}).$$

sometimes we simply write it as $s_I(c)$.

Lemma 2.6 (Thom)**.**

$$s_I(c(\omega \oplus \omega')) = \sum_{J,K} s_J(c(\omega)) s_K(c(\omega')),$$

where the summation is taken over all partitions J, K with juxtapositon JK equal to I.

Remark. We can view this formula as a generalization of the Whitney sum formula.

Proof. Let $\phi : F(B) \to B$ be the flag manifold of $\omega \oplus \omega'$, with the splitting bundle

$$\phi^*(\omega) = \eta_1 \oplus \cdots \oplus \eta_n, \quad \phi^*(\omega') = \eta'_1 \oplus \cdots \oplus \eta'_{n'}.$$

Then we have

$$\phi^* c(\omega) = (1 + c_1(\eta_1)) \cdots (1 + c_1(\eta_n)) =: (1 + t_1) \cdots (1 + t_n),$$
$$\phi^* c(\omega') = (1 + c_1(\eta'_1)) \cdots (1 + c_1(\eta'_{n'})) =: (1 + t_{n+1}) \cdots (1 + t_{n+n'}).$$

Thus

$$\begin{aligned}
\phi^* s_I(c(\omega \oplus \omega')) &= \sum t_1^{i_1} \cdots t_r^{i_r} \\
&= \sum_{\substack{JK=I,\\ J=j_1,\dots,j_{r_1},\\ K=k_1,\dots,k_{r_2}}} \left(\sum t_1^{j_1} \cdots t_{r_1}^{j_{r_1}}\right) \left(\sum t_{n+1}^{k_1} \cdots t_{n+r_2}^{k_{r_2}}\right) \\
&= \sum_{JK=I} s_J(c(\eta_1 \oplus \cdots \oplus \eta_n)) s_K(c(\eta'_1 \oplus \cdots \oplus \eta'_n)) \\
&= \sum_{JK=I} s_J(\phi^* c(\omega)) s_K(\phi^* c(\omega')) \\
&= \phi^* \left(\sum_{JK=I} s_J(c(\omega)) s_K(c(\omega')) \right).
\end{aligned}$$

Since ϕ^* is injective, we get the desired formula. □

Now for a compact complex manifold K^n. For each partition $I = i_1, \dots, i_r$ of n, we denote

$$s_I[K^n] = s_I(c)[K^n] := \langle s_I(c(\tau)), \mu_{K^n} \rangle \in \mathbb{Z}.$$

This characteristic number is clearly a linear combination of Chern numbers.

Corollary 2.7.

$$s_I[K^m \times L^n] = \sum_{JK=I} s_J[K^m] s_K[L^n].$$

where the summation is taken over all partitions J of m and K of n with juxtapositon JK equal to I.

Proof. Since $\tau_{K \times L} = \tau_K \times \tau_L = \pi_1^* \tau_K \oplus \pi_2^* \tau_L$, where π_1, π_2 are the projections from $K \times L$ to K and L respectively,

$$s_I[K \times L] = \langle s_I(c(\tau_{K \times L})), \mu_{K \times L} \rangle$$

$$
\begin{aligned}
&= \sum_{JK=I} \langle s_J(c(\pi_1^*\tau_K))s_K(c(\pi_2^*\tau_L)), \mu_K \times \mu_L\rangle \\
&= \sum_{JK=I} \langle s_J(c(\tau_K)), \mu_K\rangle\langle s_K(c(\tau_L)), \mu_L\rangle \\
&= \sum_{JK=I} s_J[K]s_K[L].
\end{aligned}
$$

□

Corollary 2.8. *As a special case,* $s_{m+n}[K^m \times L^n] = 0$*, for* $m, n > 0$*.*

There are analogous results for Pontrjagin classes and numbers. Let ξ be a real vector bundle over a compact smooth manifold B. For each $k \geq 0$ and partition $I = i_1, \ldots, i_r$ of k, we define

$$s_I(p(\xi)) := s_I(p_1(\xi), \ldots, p_k(\xi)) \in H^{4k}(B; \mathbb{Z}).$$

There are similar product formulas for Pontrjagin classes

$$s_I(p(\xi \oplus \xi')) \equiv \sum_{J,K} s_J(p(\xi))s_K(p(\xi')),$$

modulo elements of 2-torsions. Since pairing with the fundamental class annihilates all 2-torsions, we have

$$s_I[M \times N] = \sum_{JK=I} s_J[M]s_K[N],$$

Notes that these characteristic numbers will be zero unless both the dimensions of M and N are divisible by 4.

2.5 Linear Independence of Chern Numbers and of Pontrjagin Numbers

Theorem 2.9 (Thom)**.** *Let* $K^1, \ldots, K^n$ *be compact complex manifolds with* $s_k(c)[K^k] \neq 0$*. Then the* $p(n) \times p(n)$ *matrix*

$$\Big(c_{i_1} \cdots c_{i_r}[K^{j_1} \times \cdots \times K^{j_s}]\Big)$$

is nonsingular, where both $I = i_1, \ldots, i_r$ *and* $J = j_1, \ldots, j_s$ *range over all partitions of* n*.*

Similarly, we have the following theorem for Pontrjagin numbers.

Theorem 2.10 (Thom). *Let $M^4, \ldots, M^{4n}$ be compact oriented smooth manifolds with $s_k(p)[M^{4k}] \neq 0$. Then the $p(n) \times p(n)$ matrix*

$$\left(p_{i_1} \cdots p_{i_r}[M^{4j_1} \times \cdots \times M^{4j_s}]\right)$$

is nonsingular, where both $I = i_1, \ldots, i_r$ and $J = j_1, \ldots, j_s$ range over all partitions of n.

Proof. We only prove the first theorem, the second one is completely analogous.

Since the basis $c_{i_1} \cdots c_{i_r}$ isn't well behaved under the product operation, we change to the basis $s_I(c)$. By the product formula,

$$s_I[K^{j_1} \times \cdots \times K^{j_s}] \neq 0$$

only if $I = i_1, \ldots, i_r$ is a refinement of $J = j_1, \ldots, j_s$. Thus by suitable ordering of both the rows and columns, we can assume that the matrix is upper triangular. The diagonal elements are

$$s_{i_1,\ldots,i_r}[K^{i_1} \times \cdots \times K^{i_r}] = s_{i_1}[K^{i_1}] \cdots s_{i_r}[K^{i_r}] \neq 0.$$

Thus the matrix is nonsingular. $\square$

3 Thom Space & Transversality

Definition 3.1. Let ξ be a renk k vector bundle over a base space B with total space $E(\xi)$, suppose ξ is equipped with a Euclidean metric. Let

$$A = \{v \in E(\xi) \mid \|v\| > 1\}.$$

The **Thom space** of ξ is defined to be the quotient space

$$T(\xi) = E(\xi)/A.$$

Lemma 3.2. *If the base space B is a CW-complex, then the Thom space $T(\xi)$ is a $(k-1)$-connected CW-complex, having one cell of dimension $n+k$ for each cell of dimension n in B and a base point t_0.*

Remark. Notice that by the cellular approximation theorem, a CW-complex with k-skeleton trivial is $(k-1)$-connected.

Lemma 3.3. *If ξ is an orientable rank k vector bundle over B, then*

$$H_{k+i}(T(\xi), t_0; \mathbb{Z}) \cong H_i(B; \mathbb{Z}).$$

Proof.

$$\begin{aligned} H_{k+i}(T(\xi), t_0; \mathbb{Z}) &= H_{k+i}(E(\xi)/A, E_0/A; \mathbb{Z}) \\ &\cong H_{k+i}(E(\xi), E_0; \mathbb{Z}) \quad \text{by excision thm} \\ &\cong H_i(B; \mathbb{Z}) \quad \text{by Thom isom.} \end{aligned}$$

□

Theorem 3.4. *Let X be a $(k-1)$-connected finite CW-complex, $k \geq 2$. Then the Hurewicz homomorphism*

$$\pi_r(X) \to H_r(X; \mathbb{Z})$$

is a $\mathcal{C}$-isomorphism for $r < 2k-1$.

Corollary 3.5. *If T is the Thom space of an orientable rank k vector bundle over a finite CW-complex B, $k \geq 2$, then there is a $\mathcal{C}$-isomorphism*

$$\pi_{k+i}(T) \to H_i(B; \mathbb{Z})$$

for $i < k-1$.

Proof.

$$\pi_{k+i}(T) \overset{\mathcal{C}-\text{isom}}{\cong} H_{k+i}(T,t_0;\mathbb{Z}) \cong H_i(B;\mathbb{Z}), \quad i<k-1.$$

The first isomorphism holds since by Lemma 3.2 T is $(k-1)$-connected finite CW-complex, then we can apply Theorem 3.4; the second isomorphism holds since ξ is orientable, we can apply Lemma 3.3. □

Theorem 3.6. *Every continuous map $f\colon S^m \to B$ is homotopic to a map g which is smooth throughout $g^{-1}(T-t_0)$ and transverse to the zero section B. The oriented cobordism class of the submanifold $g^{-1}(B)$ depends only on the homotopy class of g. Hence there is a well-defined map*

$$\begin{aligned}\pi_m(T,t_0) &\to \Omega^{\mathrm{SO}}_{m-k}\\ [g] &\mapsto [g^{-1}(B)].\end{aligned}$$

3.1 The Main Theorem

Let $\tilde{\gamma}^k$ be the universal oriented rank k vector bundle over $\tilde{\mathrm{Gr}}_k(\mathbb{R}^\infty)$, we have the following important theorem.

Theorem 3.7 (Thom's Theorem)**.** *For $k>n+1$, the map*

$$\begin{aligned}\pi_{n+k}(T(\tilde{\gamma}^k),t_0) &\to \Omega^{\mathrm{SO}}_{n}\\ [g] &\mapsto [g^{-1}(\tilde{\mathrm{Gr}}_k(\mathbb{R}^\infty))]\end{aligned}$$

is an isomorphism. Similarly, for the unoriented case, the map

$$\begin{aligned}\pi_{n+k}(T(\tilde{\gamma}^k),t_0) &\to \Omega_{n}\\ [g] &\mapsto [g^{-1}(\tilde{\mathrm{Gr}}_k(\mathbb{R}^\infty))]\end{aligned}$$

is also an isomorphism.

Remark. 这个同构是否由这个映射给出存疑.

We will not prove this theorem, instead, we will proof part of it. Let $\tilde{\gamma}^k_p$ be the universal bundle over $\tilde{\mathrm{Gr}}_k(\mathbb{R}^{k+p})$.

Lemma 3.8. *If $k\geq n$ and $p\geq n$, then the homomorphism*

$$\begin{aligned}\pi_{n+k}(T(\tilde{\gamma}^k_p),t_0) &\to \Omega^{\mathrm{SO}}_{n}\\ [g] &\mapsto [g^{-1}(\tilde{\mathrm{Gr}}_k(\mathbb{R}^{k+p}))]\end{aligned}$$

is surjective.